Lab experiment

Crypt-Arithmetic Problem

Coding:

```
from itertools import permutations

# Crypt-Arithmetic Problem
# SEND + MORE = MONEY

letters = 'SENDMORY'

digits = range(10)

for p in permutations(digits, 8):
    S, E, N, D, M, O, R, Y = p

    # Leading digits cannot be zero
    if S == 0 or M == 0:
        continue

    SEND = 1000*S + 100*E + 10*N + D
    MORE = 1000*M + 100*O + 10*R + E
    MONEY = 10000*M + 1000*O + 100*N + 10*E + Y

    if SEND + MORE == MONEY:
        print("Solution Found:")
        print("SEND =", SEND)
        print("MORE =", MORE)
        print("MONEY =", MONEY)
        Break
```

Output:

Output

```
Solution Found:
SEND = 9567
MORE = 1085
MONEY = 10652
```